

Order Fulfillment Analytics – Business Findings

Context

This document captures business findings and analytical insights discovered during the dashboard phase. These findings will feed into the Portfolio Case Study Write-Up (Phase 6), where they will be presented as evidence-based business recommendations.

Service Performance

1. **Overall late delivery rate: 54.83%** — More than half of all orders carry a late delivery risk flag. This is a significant operational signal that requires investigation by shipping mode and market.
2. **First Class has the highest late delivery risk at 95.2%** — Counterintuitive finding. First Class is a premium faster service, yet it shows the highest risk of late delivery. Likely explanation: the tighter the service promise, the harder it is to meet consistently. Any deviation from the 2-day window gets flagged as late. Business implication: the company may be overpromising on First Class delivery. This was also supported by Phase 1 audit findings, where First Class deliveries were concentrated almost entirely in 2-day windows with very little variation.
3. **Late delivery rate by shipping mode** — First Class 95.32%, Second Class 76.63%, Same Day 45.74%, Standard Class 38.07%. The most expensive services are the least reliable. Standard Class is the most consistent performer. This pattern suggests a structural operational issue rather than isolated incidents.
4. **Average order cycle time: 3.50 days** — Consistent with Phase 1 audit findings (avg ~3.5 days). This is the baseline operational benchmark for fulfillment speed across all shipping modes and markets.
5. **Late delivery rate trend by year** — Rate rises from 2015, peaks in 2016, drops slightly in 2017, then jumps upward in 2018. However 2018 contains only Q1 data and is not representative of a full year. The 2018 spike should not be interpreted as a confirmed trend.
6. **Quarterly pattern — no clear seasonal trend confirmed** — The quarterly pattern appears to alternate between years (Q3 is the lowest point in 2016 but potentially higher in other years), suggesting no consistent seasonal cycle. Two to three years of data is insufficient to confirm a reliable pattern. This finding should be noted with caution in the case study.

Profitability

7. **Overall profit margin: 10.78%** — In line with typical retail/supply chain margins (5–15% range).
8. **Shipping Cost Ratio not available** — The dataset does not capture shipping cost. This planned KPI from the project charter could not be delivered as scoped; a Discount vs Profit Margin analysis was used instead (see below).
9. **Maximum discount rate: 25%** — No orders in the dataset exceed a 25% discount. This suggests an effective cap on discount policy, whether by company rule or by what this dataset captures.
10. **Profit margin decreases as discount rate increases** — Profit margin drops steadily from the 0–5% discount band (highest margin) through the 15–20% band (lowest in the main trend), confirming the expected business relationship: deeper discounts directly erode profitability.
11. **20–25% discount band shows an unexpected margin uptick — likely a sample-size artifact** — This band breaks the otherwise clean decreasing trend. Investigation showed the 20–25% bins contain only 10,029 orders each, roughly five times fewer than the 0–5% band (50,142 orders). With a much smaller sample, the average margin is more sensitive to outliers. Conclusion: the core discount-erodes-margin trend is reliable across the bulk of the data (0–20%, large sample sizes); the high-discount uptick should not be presented as a confirmed reversal without further investigation.

Risk

12. **Late delivery risk does not vary meaningfully by customer segment** — Consumer, Corporate and Home Office all sit within a 54.7–55.1% late delivery rate — a 0.4 percentage point spread, effectively flat. The real differentiator between segments is sales volume, not risk: Consumer drives ~19M in sales, Corporate ~10M, Home Office ~6M.
13. **Late delivery risk also does not vary meaningfully by market** — Same pattern as customer segment: late delivery rate is essentially flat across Europe, LATAM, USCA, Pacific Asia and Africa. Africa stands out only for having the lowest sales volume of all markets, not for elevated risk.
14. **Region-level risk (order_region) shows more apparent variation — unconfirmed, needs validation** — At the finer order_region grain, late delivery rate swings between roughly 48% and 56%, with the sharpest spikes around the smallest-volume regions (Canada, Central Asia). Given the discount_band precedent, this is likely at least partly a small-sample effect rather than genuine regional risk variation. Not used as a confirmed finding — would require an order-count check per region before drawing conclusions.
15. **Perfect Order Rate not available** — The dataset only captures one dimension of order quality (on-time delivery). A genuine Perfect Order Rate also requires data on order completeness, damage and documentation accuracy, none of which exist here.

This planned KPI from the project charter was dropped rather than approximated, to avoid presenting a duplicate of OTIF under a misleading name.

Python EDA — Additional Findings

16. **33,784 orders (18.7%) are loss-making** — Not visible in the Power BI dashboard, where profit margin was shown only as an aggregate (10.78%). Python EDA revealed the distribution underneath: nearly 1 in 5 orders loses money. Discount rate does not explain this — loss-making orders have the same mean discount rate (10.2%) as the full dataset (10.1%), and some orders lose money with zero discount applied.
17. **Delivery times are bimodal — the average of 3.50 days is misleading** — The histogram reveals a spike at exactly 2 days (~56,000 order items, driven by First Class) and a flat plateau from 3–6 days (~29,000 order items each for the remaining modes). The average hides two completely different operational profiles coexisting in the same operation.
18. **Standard Class and Second Class are operationally identical** — The boxplot shows their delivery time distributions are virtually indistinguishable (both spreading from 2–6 days, median ~4 days). The four-tier shipping structure reduces to two real operational tiers: express (First Class, exactly 2 days) and standard (everything else, 3–6 days).
19. **Discount-profit correlation is near-zero at row level but negative in aggregate** — Individual correlation (-0.003), correlation heatmap, and OLS regression all confirm no detectable linear relationship at the order item level. The negative trend visible in Power BI (margin decreases per discount band) only emerges when data is grouped and averaged — at the row level, too many other factors dominate. This aggregation-level nuance should be preserved in the case study.